Mask

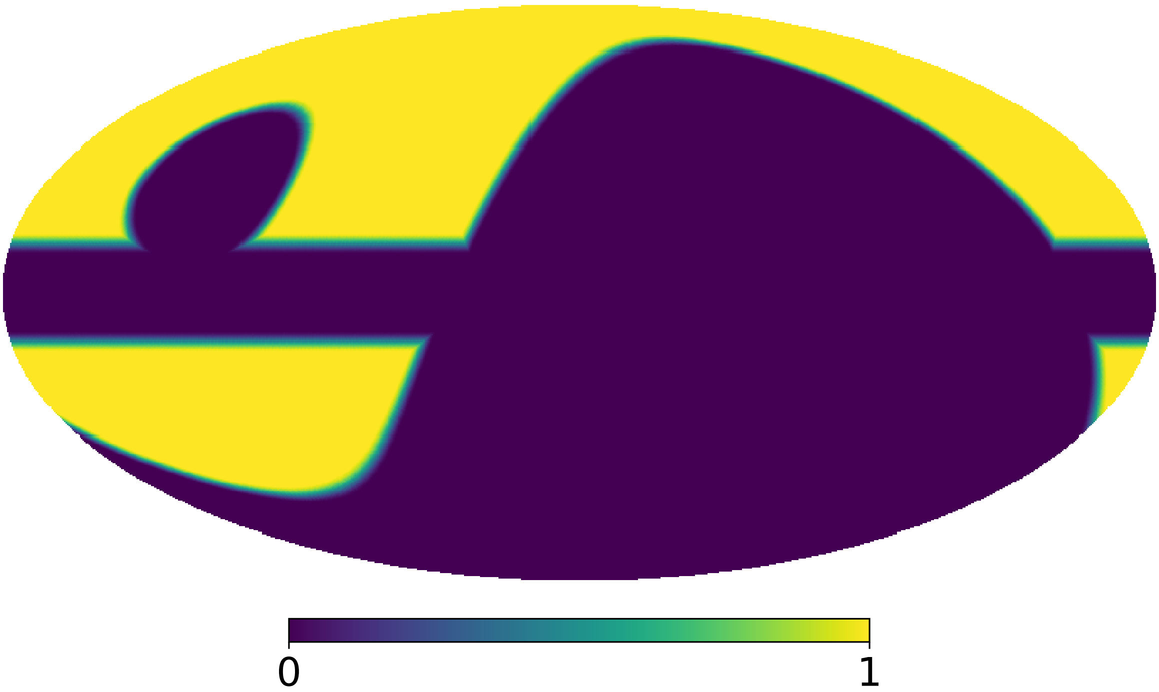

Figura 1: Mask used for the analysis.

MCMC Fitting

The total angular power spectrum between frequencies ν_i and ν_j is modeled as the sum of synchrotron, dust, and synchrotron–dust cross-correlation components,

$$C_\ell(\nu_i, \nu_j) = C_\ell^{\rm sync}(\nu_i, \nu_j) + C_\ell^{\rm dust}(\nu_i, \nu_j) + C_\ell^{\rm sd}(\nu_i, \nu_j), \tag{1}$$

where the synchrotron term is

$$C_\ell^{\rm sync}(\nu_i, \nu_j) = A_s \left(\frac{\ell}{\ell_{\rm ref}}\right)^{\alpha_s} \left(\frac{\nu_i}{\nu_{\rm ref}}\right)^{\beta_s} \left(\frac{\nu_j}{\nu_{\rm ref}}\right)^{\beta_s}, \tag{2}$$

the dust term is

$$C_\ell^{\rm dust}(\nu_i, \nu_j) = A_d \left(\frac{\ell}{\ell_{\rm ref}}\right)^{\alpha_d} S(\nu_i, \nu_0, \beta_d, T_d)\, S(\nu_j, \nu_0, \beta_d, T_d), \tag{3}$$

with the modified blackbody scaling

$$S(f, \nu_0, \beta_d, T_d) = \frac{B_\nu(f, T_d)}{B_\nu(\nu_0, T_d)} \left(\frac{\nu}{\nu_0}\right)^{\beta_d}, \tag{4}$$

and the cross-correlation term is

$$C_\ell^{\rm sd}(\nu_i, \nu_j) = \rho\sqrt{A_s A_d}\,[S_s(\nu_i)S_d(\nu_j) + S_s(\nu_j)S_d(\nu_i)] \left(\frac{\ell}{\ell_{\rm ref}}\right)^{(\alpha_s+\alpha_d)/2}, \tag{5}$$

where $S_s(f) = (\nu/\nu_{\mathrm{ref}})^{\beta_s}$ and $S_d(\nu) = S(\nu, \nu_0, \beta_d, T_d)$ denote the synchrotron and dust frequency scalings, respectively. Here, A_s and A_d are the amplitudes of the synchrotron and dust components at the reference multipole ℓ_{ref} (taken as 80) and frequencies ν_{ref} (11.1 GHz) and ν_0 (353.0 GHz), α_s and α_d are the multipole spectral indices, β_s and β_d the frequency spectral indices and T_d the dust temperature.

We perform a Markov Chain Monte Carlo (MCMC) fitting procedure to model the total angular power spectrum across multiple frequency bands, using the synchrotron, dust, and synchrotron–dust cross-correlation model described above. Specifically, we fit simulated sky maps from the following frequency bands: QUIJOTE 11 GHz; WMAP 23, 33, 41, 61, and 94 GHz; and Planck 30, 44, 70, 100, 143, 217, and 353 GHz. These maps are generated using the `Python` library `PySM`, including all the radiation components of the sky. These maps have been smoothed using the beam window functions provided by each experiment. The resulting posterior distributions are visualized in the corner plots: Figure 2 for the EE mode and Figure 3 for the BB mode, showing the constraints on the synchrotron and dust parameters and their correlations.

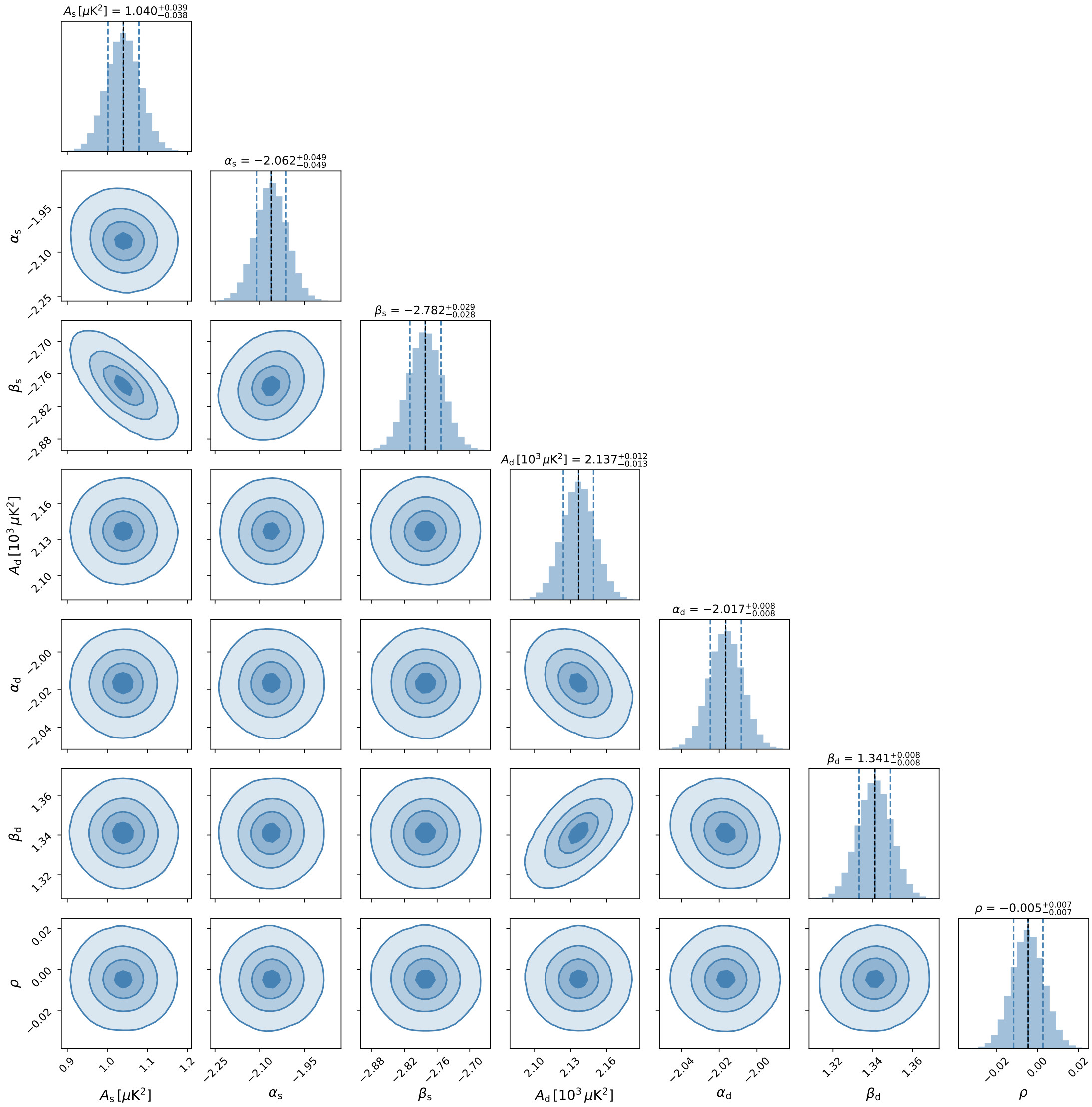

Figura 2: Corner plot of the posterior distributions for the synchrotron and dust model parameters, accounting for correlations, obtained from the EE mode data.

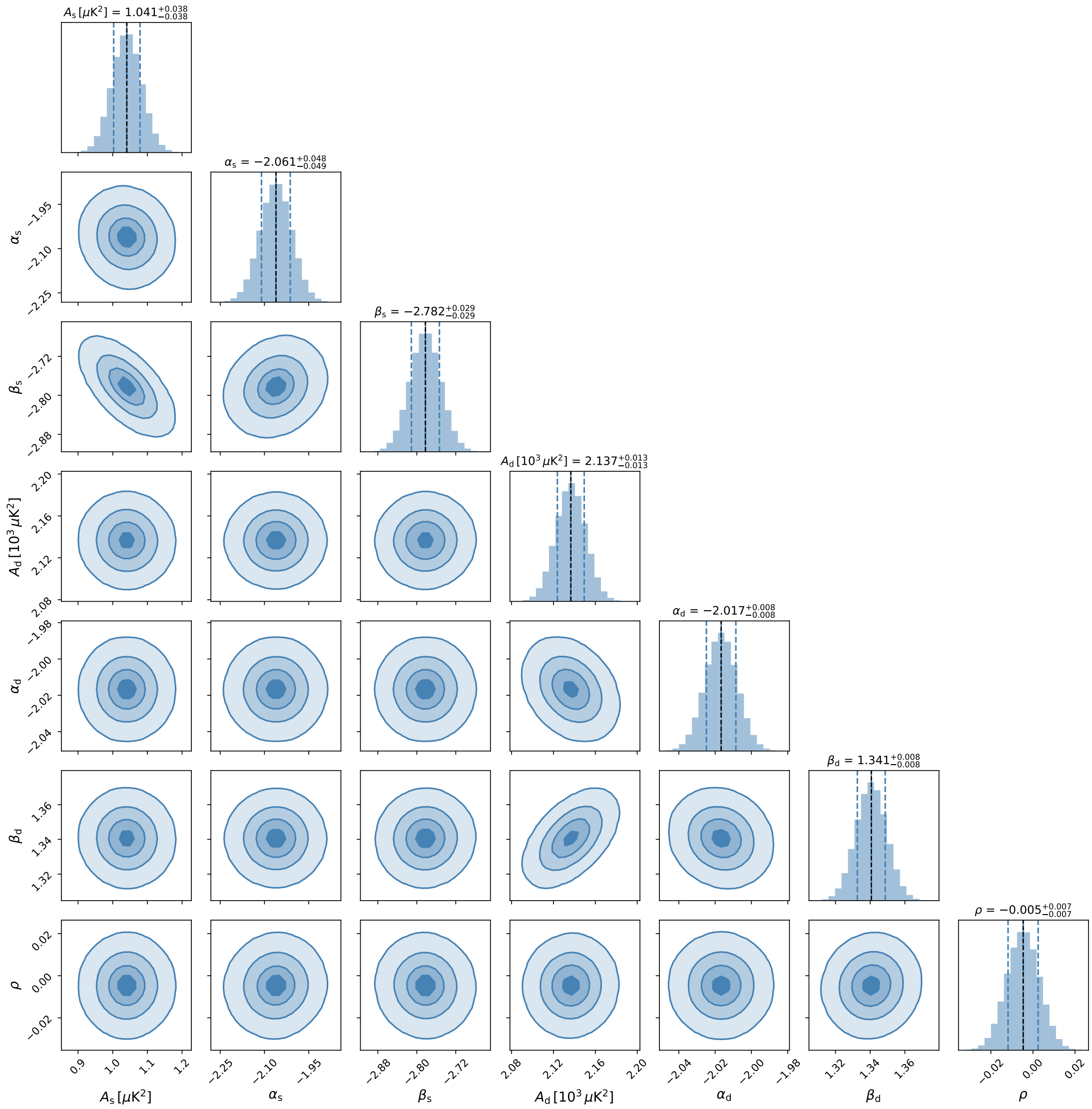

Figura 3: Corner plot of the posterior distributions for the synchrotron and dust model parameters, accounting for correlations, obtained from the BB mode data.